
ECSE 4660/6660: Internetworking of Things

Lecture 9: Link Layer Fundamentals – Part I: MAC protocols

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Administrative

- Use Piazza for any questions
 - <https://piazza.com/rpi/spring2026/ece46606660>
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Lab
 - TA hours 4-5 PM daily in 5312 lab
 - Lab 2 due Fri, 2/13
- HW2 soon on **LMS** due 2/20
 - No Late submission allowed
- Project group formation
 - [IoT_SP26_project.xlsx](#)
- Project Next Steps
 - Work with TA to get the hardware needed
 - Mid-term report due 3/20
 - In-class presentation+2min video 4/20, 4/23
 - Live-demo 4/27
 - Final code/report 5/1

Link Layer Techniques: Medium Access (4 Lectures)

- MAC protocols: Aloha, CSMA
 - RFID, LoRa: Aloha, Slotted-Aloha
 - Wired Network: CSMA/CD
 - IEEE 802.11 b/g/n/ac WiFi: CSMA/CA
 - Cellular MAC (NB-IoT): CDMA, TDMA, FDMA, OFDMA
- Other IoT Protocols
 - Bluetooth Low Energy (BLE)
 - IEEE 802.15.4 and Zigbee

References

Link Layer Fundamentals: Reference for this part:

- A. Kumar, D. Manjunath, J. Kuri, “Communication Networking, An Analytical Approach,” 2004, Elsevier. (Available as **E-BOOK** through the RPI Library)

Chapter 8: Sections:

8.5.2: Carrier Sensing Protocols

8.6.1: The IEEE 802.11 Standard

8.6.2: Multiple Access in IEEE 802.11

- J.F. Kurose, K.W. Ross, “*Computer Networking, A Top-Down Approach Featuring the Internet*” (7th Ed.).

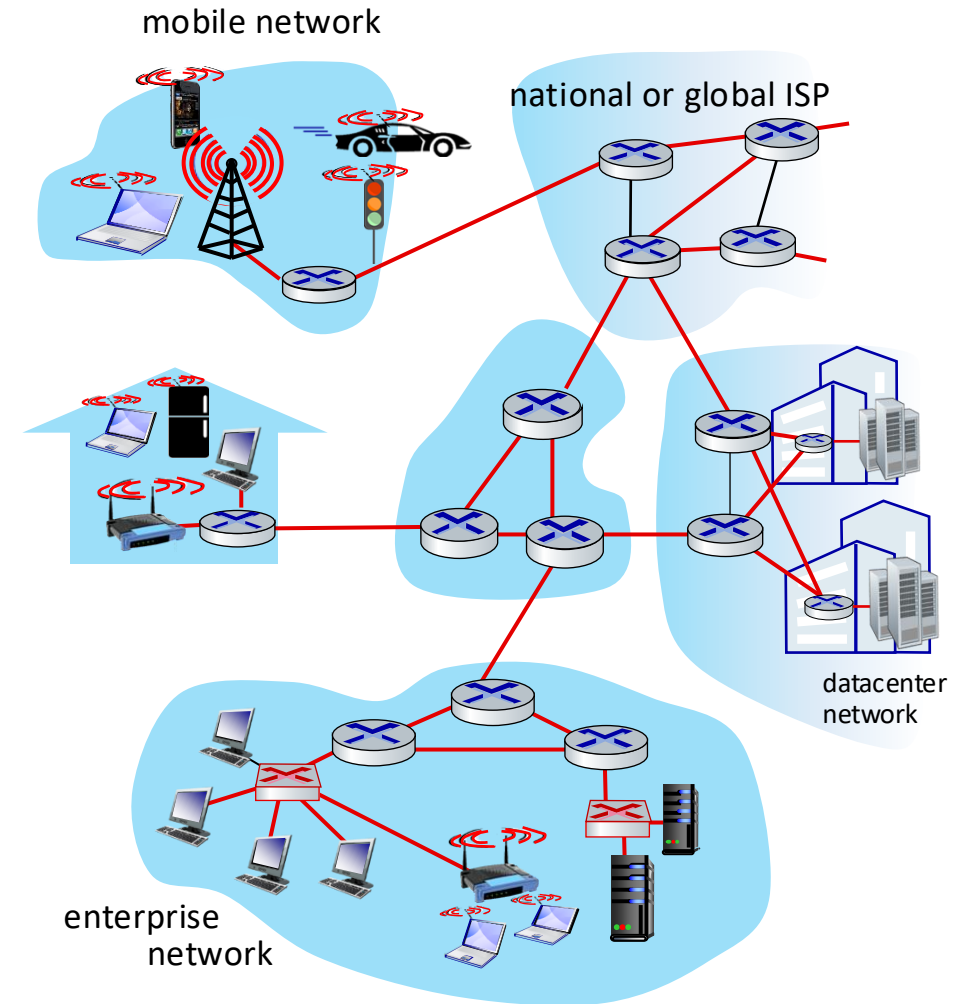
Chapter 7: Wireless Networks (posted on Piazza)

Link layer: introduction

terminology:

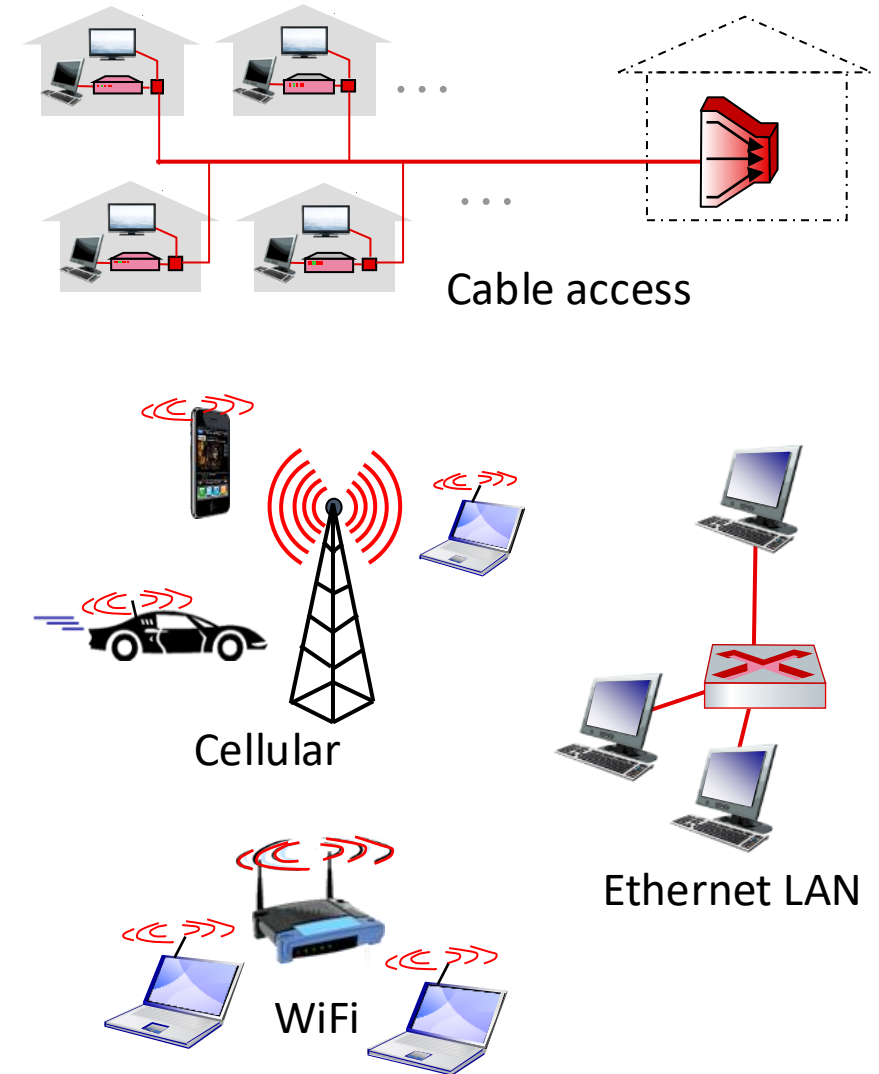
- hosts and routers: nodes
- communication channels that connect adjacent nodes along communication path: links
 - wired
 - wireless
 - LANs
- layer-2 packet: *frame*, encapsulates datagram

link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link



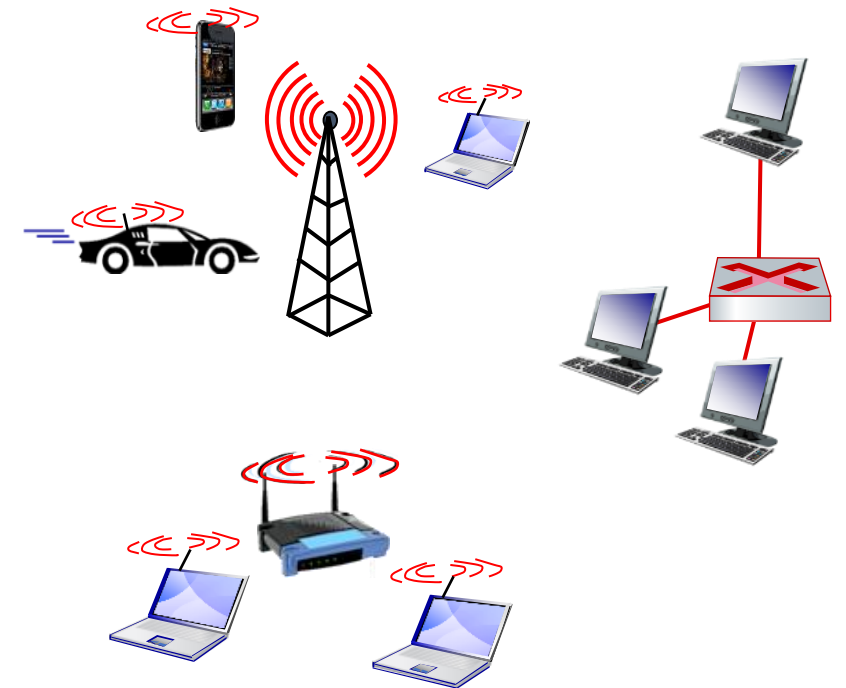
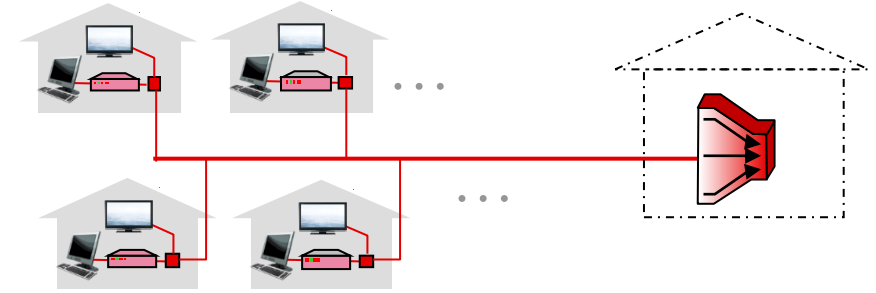
Link layer: services

- **framing, link access:**
 - encapsulate datagram into frame, adding header, trailer
 - channel access if shared medium
 - “MAC” addresses in frame headers identify source, destination (different from IP address!)
- **reliable delivery between adjacent nodes**
 - we already know how to do this!
 - seldom used on low bit-error links
 - wireless links: high error rates
 - Q: why both link-level and end-end reliability?



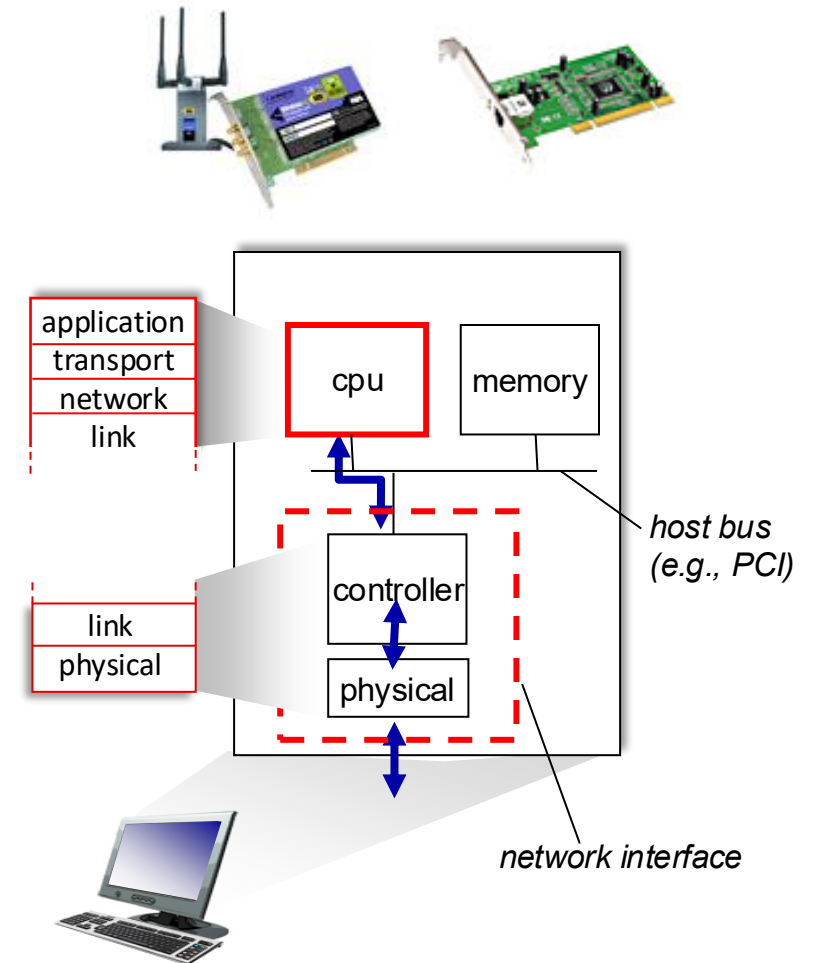
Link layer: services (more)

- **flow control:**
 - pacing between adjacent sending and receiving nodes
- **error detection:**
 - errors caused by signal attenuation, noise.
 - receiver detects errors, signals retransmission, or drops frame
- **error correction:**
 - receiver identifies *and corrects* bit error(s) without retransmission
- **half-duplex and full-duplex:**
 - with half duplex, nodes at both ends of link can transmit, but not at same time

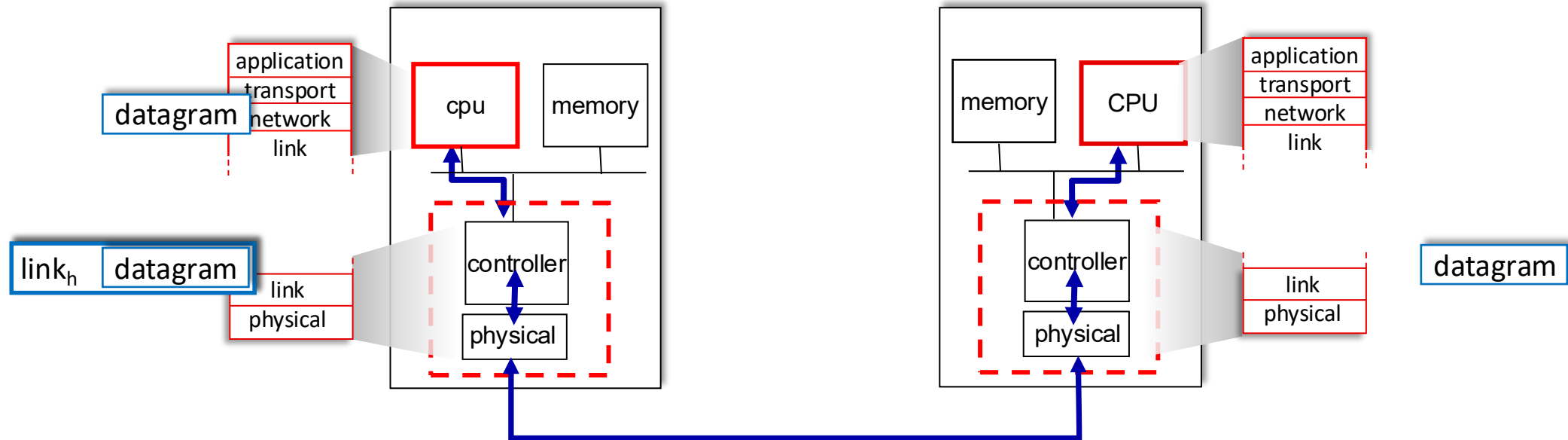


Where is the link layer implemented?

- in each-and-every host
- link layer implemented in *network interface card* (NIC) or on a chip
 - Ethernet, WiFi card or chip
 - implements link, physical layer
- attaches into host's system buses
- combination of hardware, software, firmware



Interfaces communicating



sending side:

- encapsulates datagram in frame
- adds error checking bits, reliable data transfer, flow control, etc.

receiving side:

- looks for errors, reliable data transfer, flow control, etc.
- extracts datagram, passes to upper layer at receiving side

Multiple access links, protocols

two types of “links”:

- point-to-point
 - point-to-point link between Ethernet switch, host
 - PPP for dial-up access
- **broadcast (shared wire or medium)**
 - old-fashioned Ethernet
 - upstream HFC in cable-based access network
 - 802.11 wireless LAN, 4G/4G. satellite



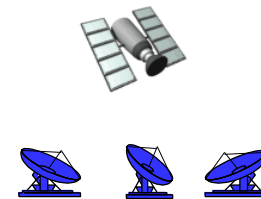
shared wire (e.g.,
cabled Ethernet)



shared radio: 4G/5G



shared radio: WiFi



shared radio: satellite



humans at a cocktail party
(shared air, acoustical)

Media Access Control (MAC) protocols

- single shared broadcast channel
- two or more simultaneous transmissions by nodes: interference
 - *collision* if node receives two or more signals at the same time

MAC protocol

- distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit
- communication about channel sharing must use channel itself!
 - no out-of-band channel for coordination

An ideal multiple access protocol

given: multiple access channel of rate R bps

requirements:

1. when one node wants to transmit, it can send at rate R .
2. when M nodes want to transmit, each can send at average rate R/M
3. fully decentralized:
 - no special node to coordinate transmissions
 - no synchronization of clocks, slots
4. simple

MAC protocols: taxonomy

two broad classes:

- **channel partitioning**

- divide channel into smaller “pieces” (time slots, frequency, code)
- allocate piece to node for exclusive use

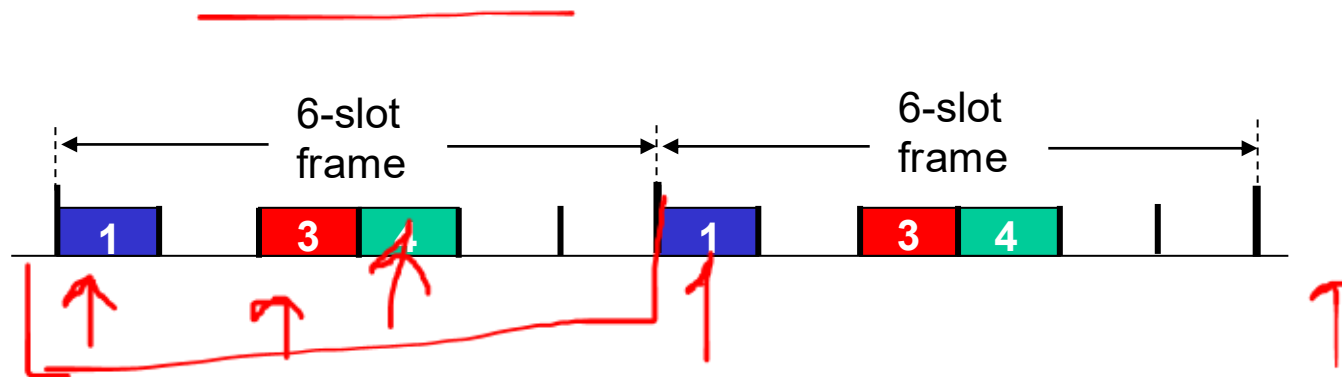
- ***random access***

- channel not divided, allow collisions
- “recover” from collisions

Channel partitioning MAC protocols: TDMA

TDMA: time division multiple access

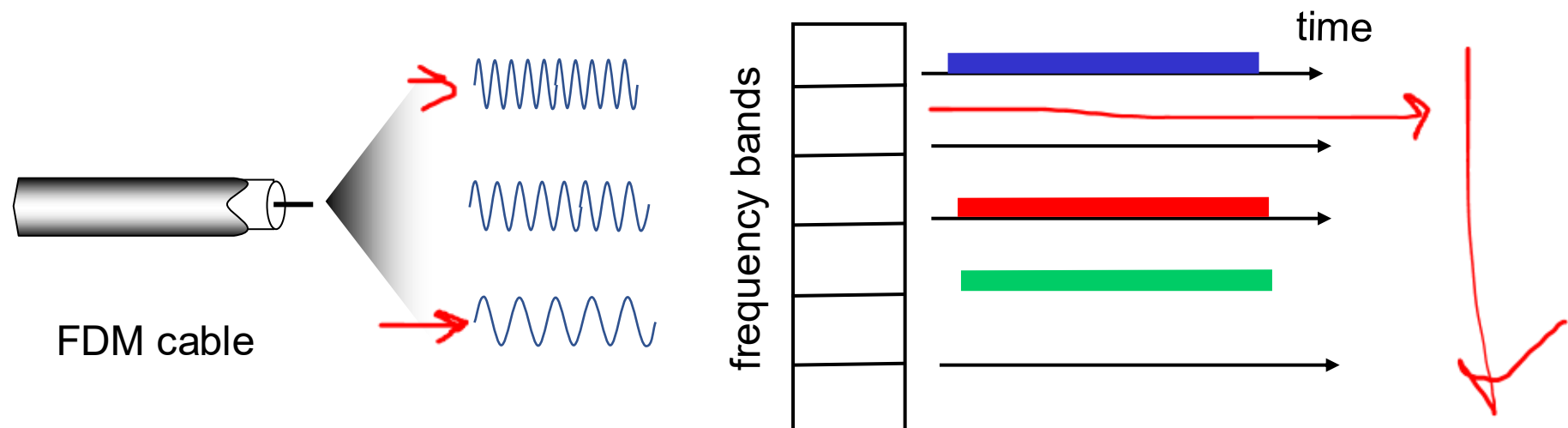
- access to channel in “rounds”
- each station gets fixed length slot (length = packet transmission time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have packets to send, slots 2,5,6 idle



Channel partitioning MAC protocols: FDMA

FDMA: frequency division multiple access

- channel spectrum divided into frequency bands
- each station assigned fixed frequency band
- unused transmission time in frequency bands go idle
- example: 6-station LAN, 1,3,4 have packet to send, frequency bands 2,5,6 idle



Random access protocols

- when node has packet to send
 - transmit at full channel data rate R .
 - no *a priori* coordination among nodes
- ~~two or more transmitting nodes: “collision”~~
- random access MAC protocol specifies:
 - when to send
 - how to detect collisions
 - how to recover from collisions (e.g., via delayed retransmissions)
- examples of random access MAC protocols:
 - ✓ ALOHA, slotted ALOHA
 - ✓ CSMA, CSMA/CD, CSMA/CA

ALOHA

ALOHAnet (1971)

- University of Hawaii – Norman Abramson
- First demonstration of wireless packet network

Rules

1. If you have data to send, send it

2. Two (or more) simultaneous transmissions will collide and be lost
- ✓ • Wait a duration of time for an acknowledgement
 - ✓ • If transmission was lost, try sending again “later”
 - Want some kind of exponential backoff scheme with entropy here

Slotted ALOHA

assumptions:

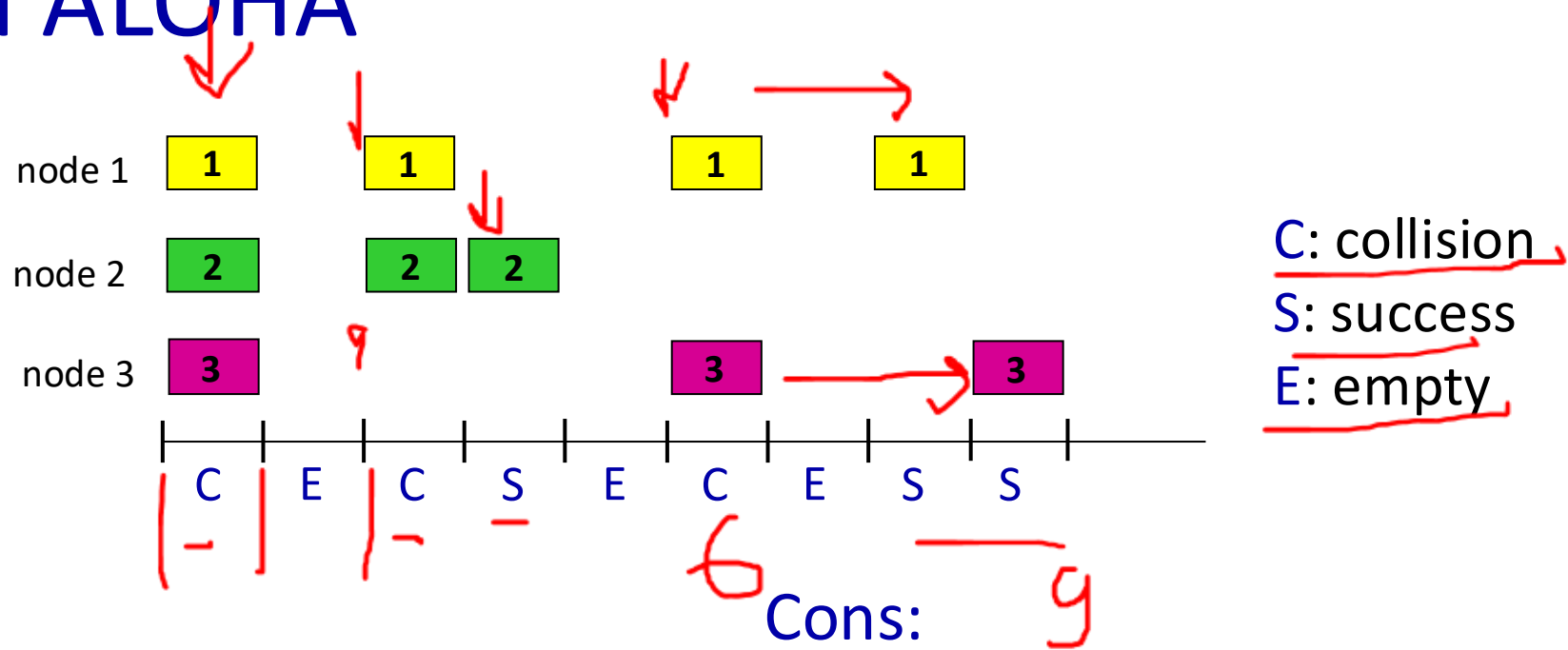
- all frames same size
- time divided into equal size slots (time to transmit 1 frame)
- nodes start to transmit only slot beginning
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision

operation:

- when node obtains fresh frame, transmits in next slot
 - *if no collision*: node can send new frame in next slot
 - *if collision*: node retransmits frame in each subsequent slot with probability p until success

randomization – why?

Slotted ALOHA



Pros:

- single active node can continuously transmit at full rate of channel
- highly decentralized: only slots in nodes need to be in sync
- simple

Cons:

- synchronization
- collisions, "wasting slots"
- idle slots, "wasting slots"

Slotted ALOHA: efficiency

efficiency: long-run fraction of successful slots (many nodes, all with many frames to send)

- *suppose:* N nodes with many frames to send, each transmits in slot with probability p

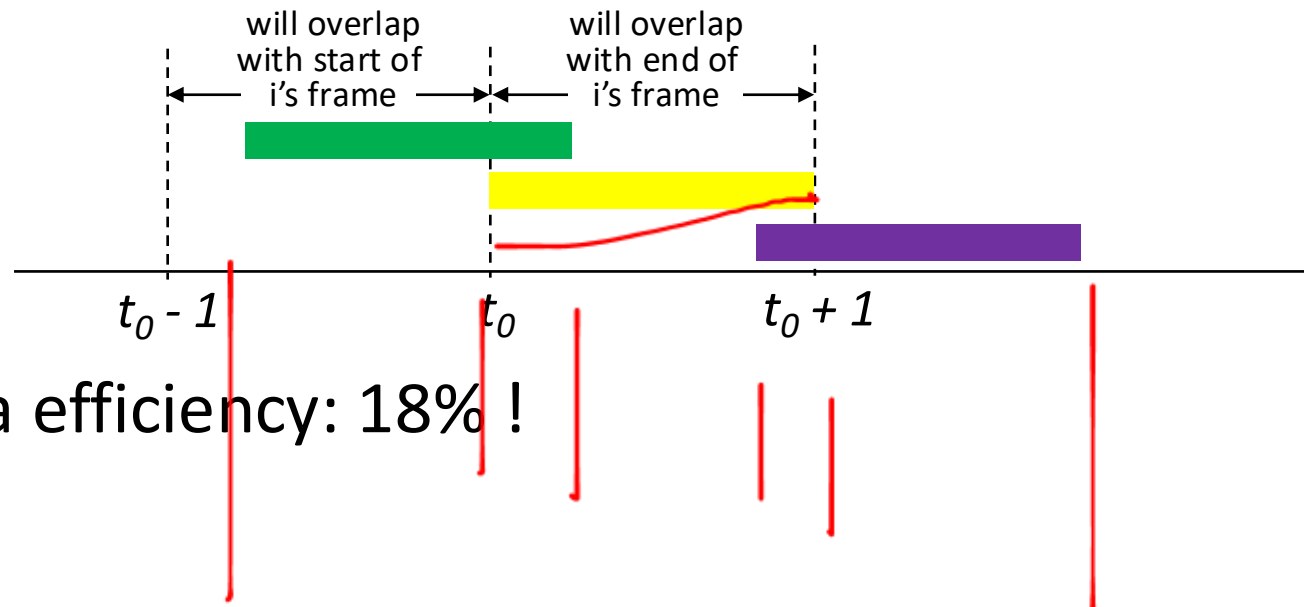
- prob that given node has success in a slot = $p(1-p)^{N-1}$
- prob that any node has a success = $Np(1-p)^{N-1}$
- max efficiency: find p^* that maximizes $Np(1-p)^{N-1}$
- for many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as N goes to infinity, gives:

$$\text{max efficiency} = 1/e = .37$$

- *at best:* channel used for useful transmissions 37% of time!

Pure ALOHA

- unslotted Aloha: simpler, no synchronization
 - when frame first arrives: transmit immediately
- collision probability increases with no synchronization:
 - frame sent at t_0 collides with other frames sent in $[t_0-1, t_0+1]$



- pure Aloha efficiency: 18% !

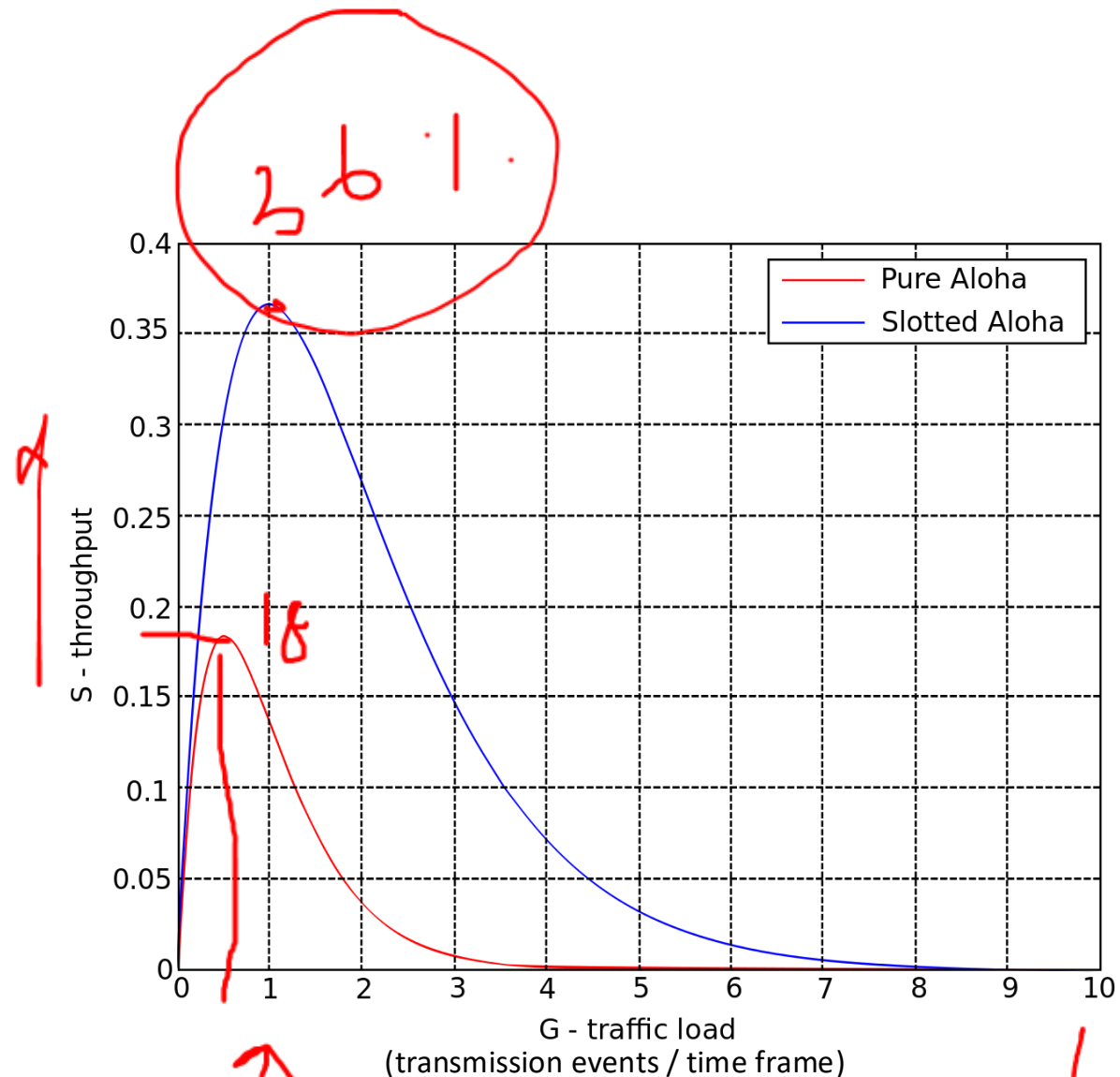
ALOHA throughput

It can be shown that traffic maxes out at

- ALOHA: 18.4%
- Slotted ALOHA: 36.8%

Assuming Poisson distribution of transmission attempts

Slotted throughput is double because the “before” collisions can no longer occur



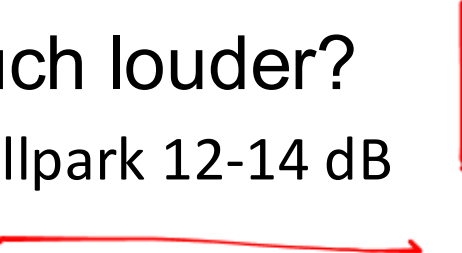
Capture effect



Actually, two packets at once isn't *always* a total loss

- The louder packet can still sometimes be heard if loud enough

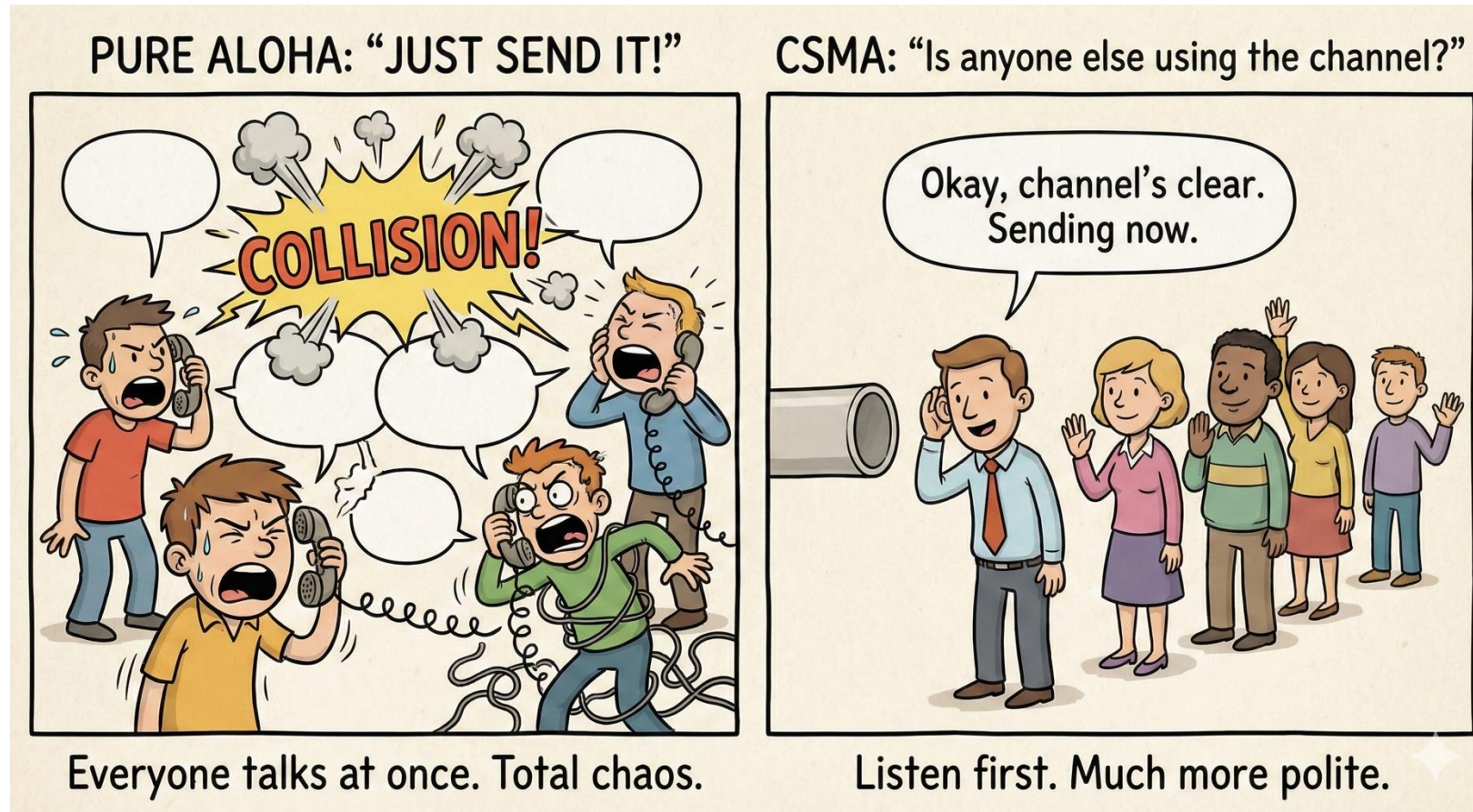
How much louder?

- Ballpark 12-14 dB
- 

When does this work?

- Depends on the radio hardware
- Louder packet first almost always works
- Louder packet second *sometimes* works

CSMA (carrier sense multiple access)



CSMA (carrier sense multiple access)

simple **CSMA**: listen before transmit:

- if channel sensed idle: transmit entire frame
 - if channel sensed busy: defer transmission
- human analogy: don't interrupt others!

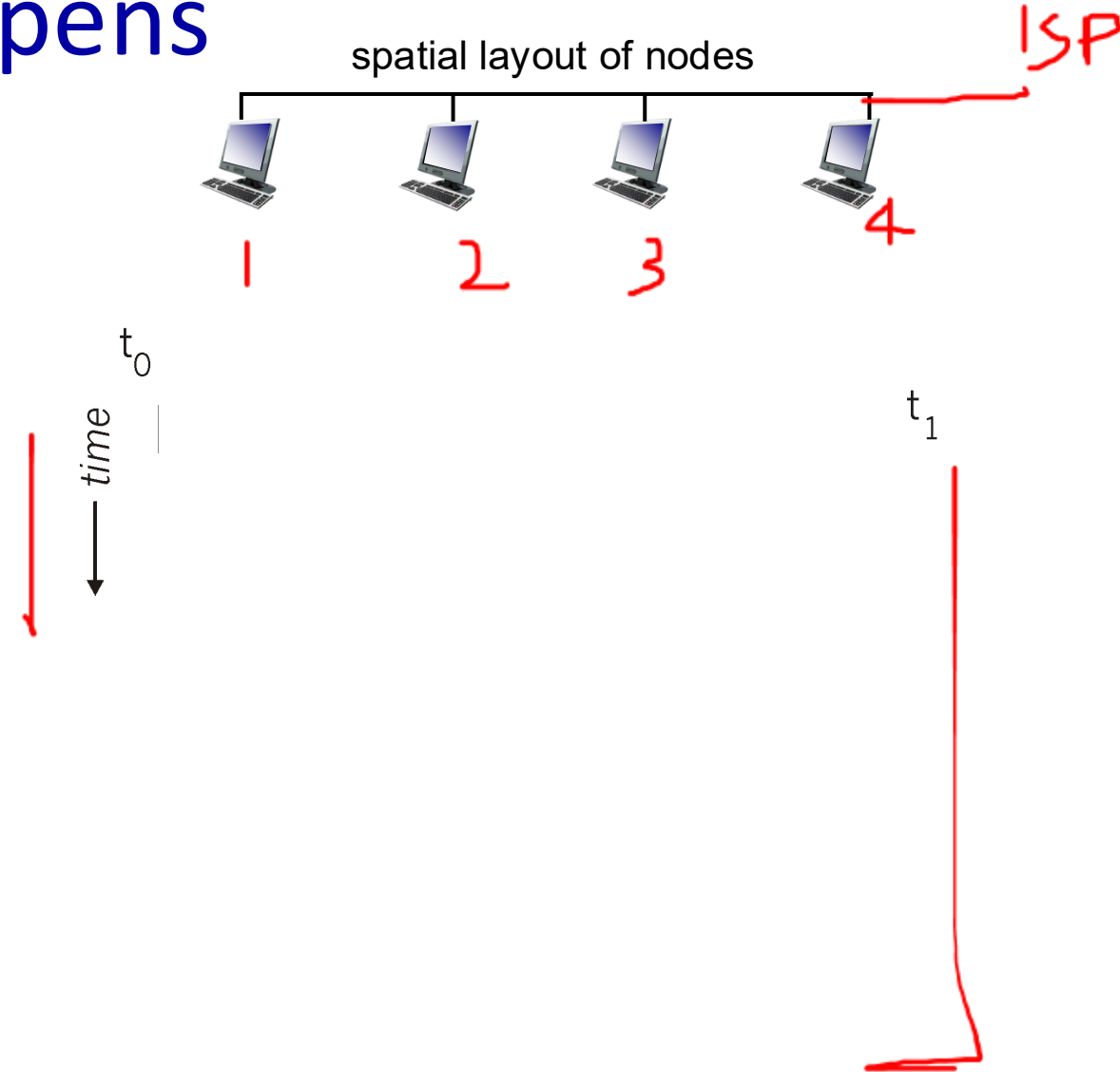
CSMA/CD: CSMA with collision detection

- collisions *detected* within short time
 - colliding transmissions aborted, reducing channel wastage
 - collision detection easy in wired, difficult with wireless
- human analogy: the polite conversationalist

CSMA/CA: CSMA with *collision avoidance*
collisions *avoidance* in wireless networks

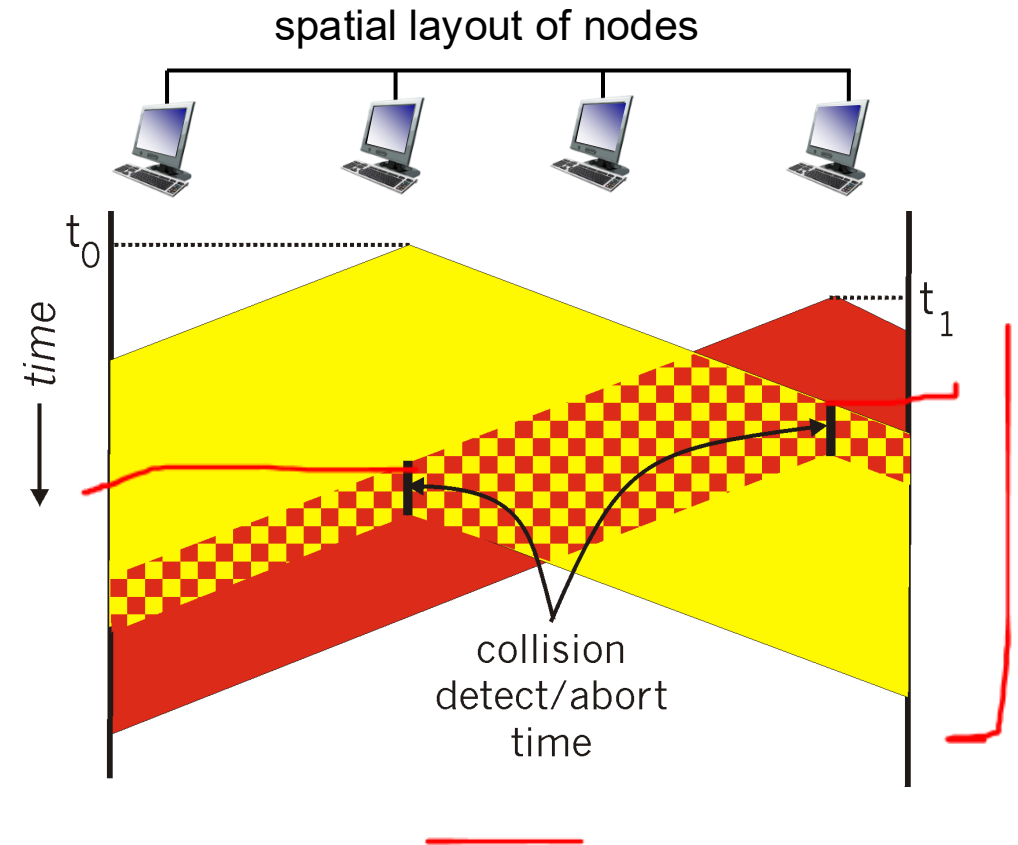
CSMA: collisions still happens

- collisions *can* still occur with carrier sensing:
 - propagation delay means two nodes may not hear each other's just-started transmission
- **collision**: entire packet transmission time wasted
 - distance & propagation delay play role in determining collision probability



CSMA/CD:

- CSMA/CS reduces the amount of time wasted in collisions
 - transmission aborted on collision detection



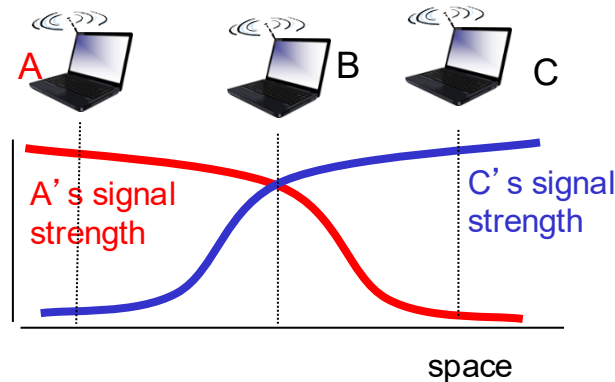
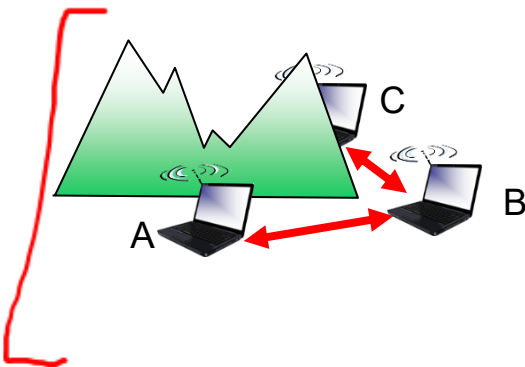
Ethernet CSMA/CD algorithm

1. Ethernet receives datagram from network layer, creates frame
2. If Ethernet senses channel:
 - if **idle**: start frame transmission.
 - if **busy**: wait until channel idle, then transmit
3. If entire frame transmitted without collision – done!
4. If another transmission detected while sending: abort, send jam signal
5. After aborting, enters binary (exponential) backoff:
 - after m th collision, chooses K at random from $\{0, 1, 2, \dots, 2^m - 1\}$. NIC waits $K \cdot 512$ bit times, returns to Step 2
 - more collisions: longer backoff interval

$\{0, 1\}$
 $\{0, 1, 2, 3\}$

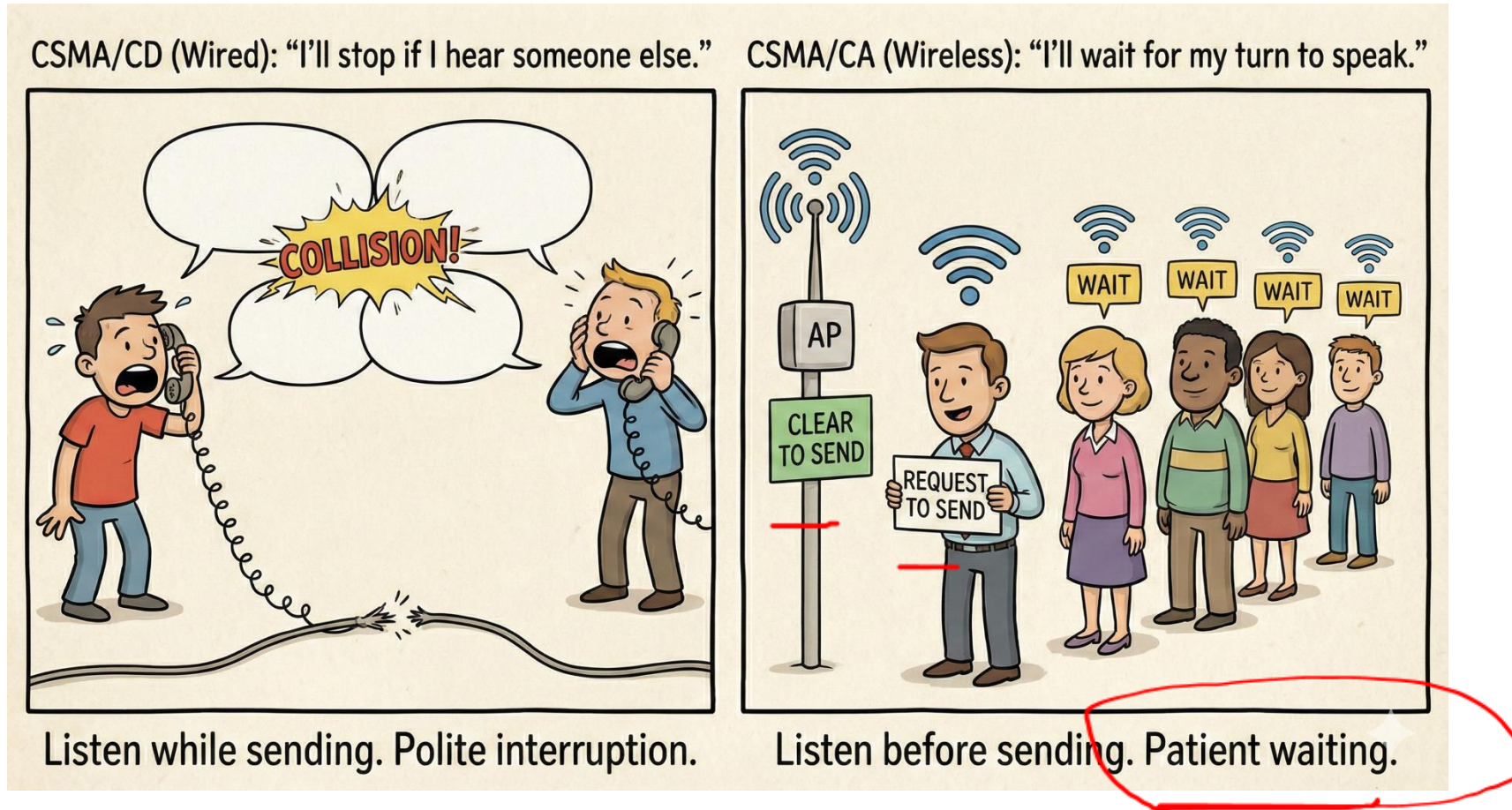
WiFi - IEEE 802.11 MAC Protocol: CSMA/CA

- avoid collisions: 2⁺ nodes transmitting at same time
- ~~802.11: CSMA - sense before transmitting~~
 - don't collide with detected ongoing transmission by another node
- 802.11: *no* collision detection (no CD)!
 - difficult to sense collisions: Radios are NOT full-duplex
 - high transmitting signal, weak received signal due to fading
 - can't sense all collisions in any case: hidden terminal, fading
 - goal: *avoid collisions*: CSMA/CollisionAvoidance



Hidden terminal problem!
A and C cannot listen to each other or "hidden" from each other

WiFi - IEEE 802.11 MAC Protocol: CSMA/CA



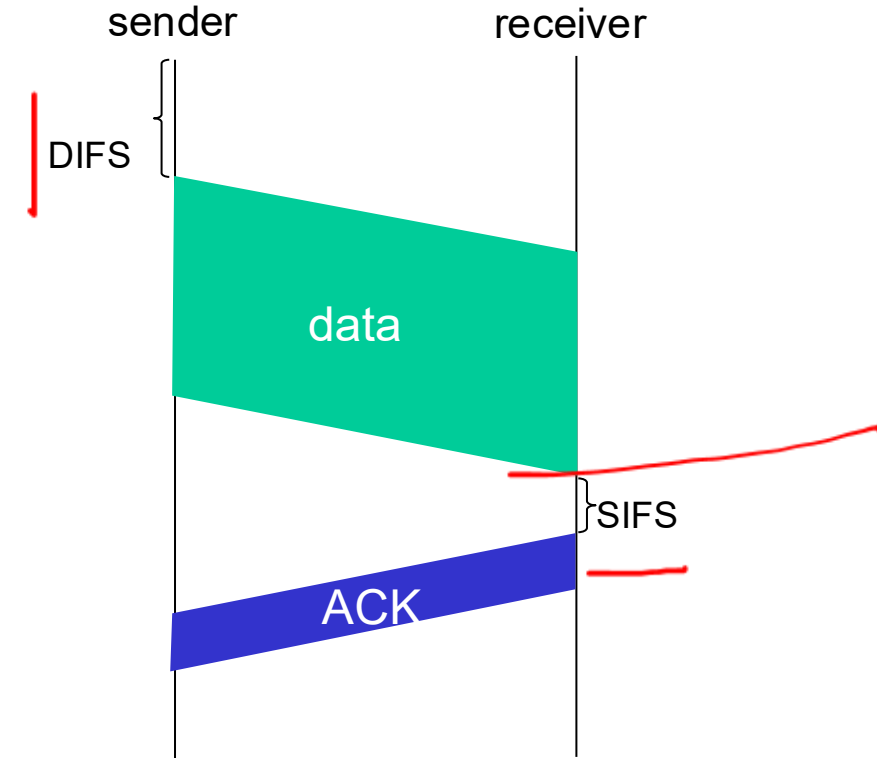
WiFi - IEEE 802.11 MAC Protocol: CSMA/CA

802.11 sender

- 1 if sense channel idle for **DIFS** then
transmit entire frame (no CD)
- 2 if sense channel busy then
 - start random backoff time
 - timer counts down while channel idle
 - transmit when timer expires
 - if no ACK, increase random backoff interval, repeat 2

802.11 receiver

if frame received OK
return ACK after **SIFS** (ACK needed due to hidden terminal problem)



Q: Why SIFS is shorter duration than DIFS?

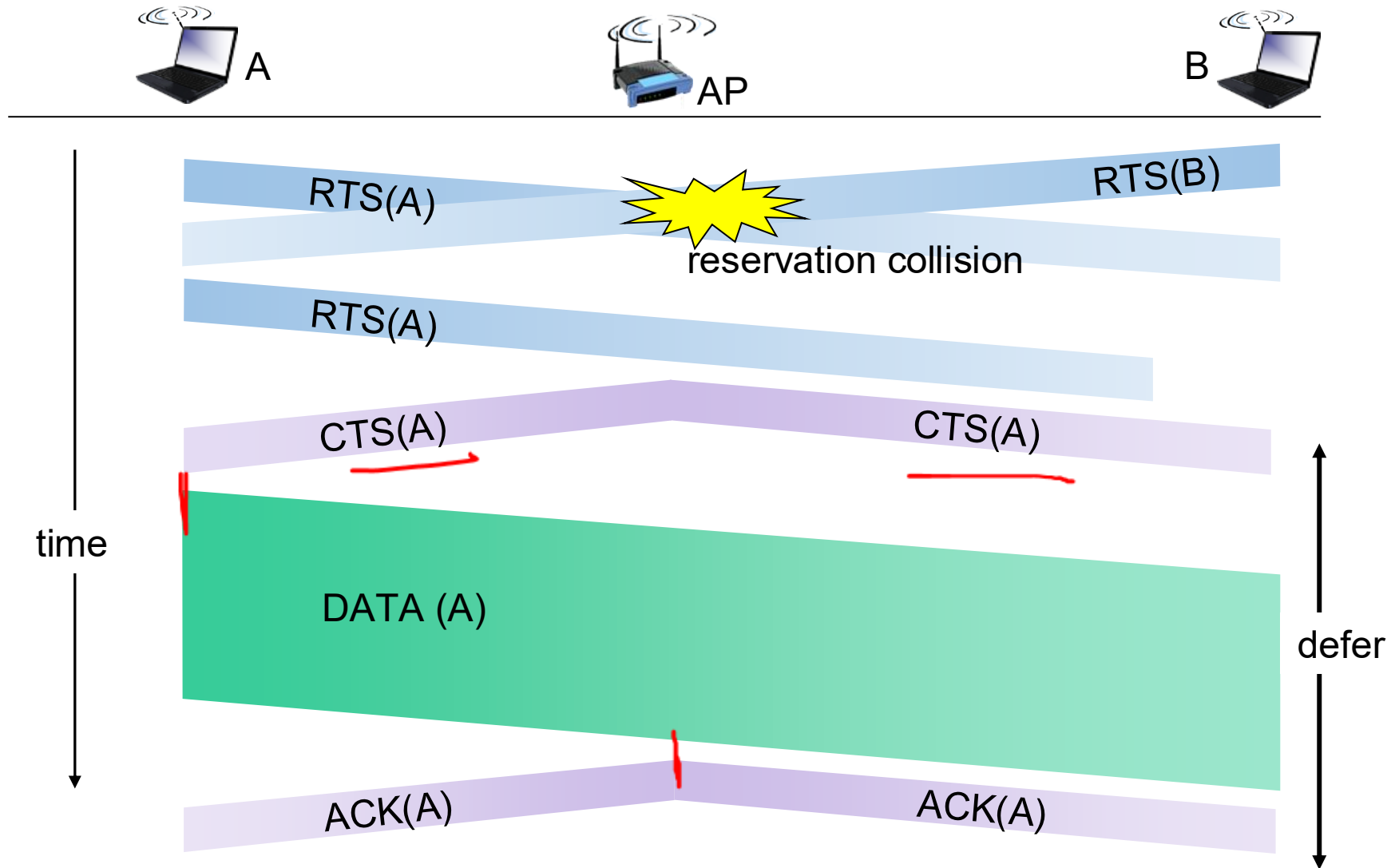
DIFS (Distributed Inter-Frame Space), SIFS (Short Inter-Frame Space)

Avoiding collisions (hidden terminal problem)

idea: sender “reserves” channel use for data frames using small reservation packets

- sender first transmits *small* request-to-send (RTS) packet to BS using CSMA
 - RTSs may still collide with each other (but they’re short)
- BS broadcasts clear-to-send (CTS) in response to RTS
- CTS heard by all nodes
 - sender transmits data frame
 - other stations defer transmissions

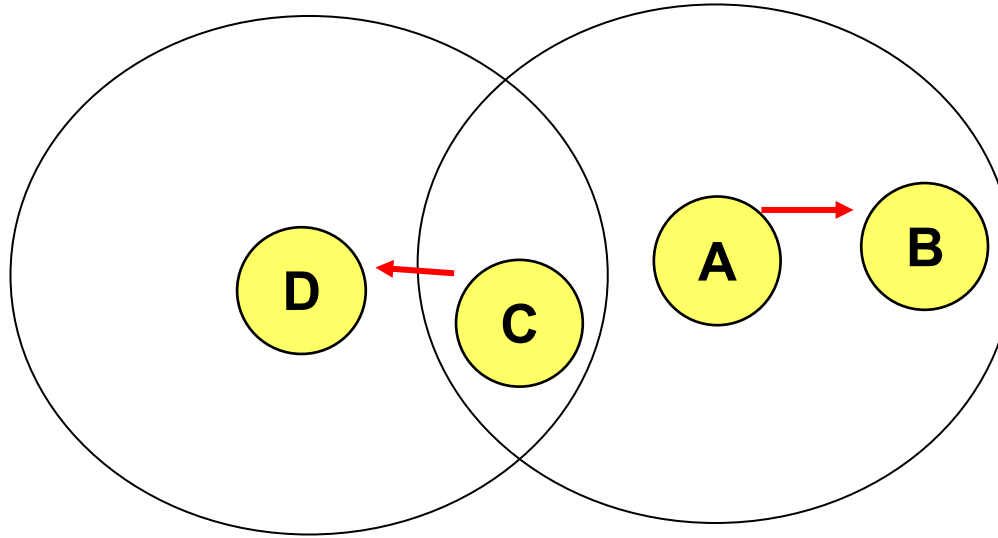
Collision Avoidance: RTS-CTS exchange



Collision Avoidance: RTS-CTS exchange

- CSMA/CA alone does not effectively solve the hidden terminal problem because it relies on local sensing of the medium, which is insufficient when devices are out of range of each other.
- The RTS/CTS mechanism complements CSMA/CA and successfully addresses the hidden terminal problem by explicitly coordinating transmissions and preventing collisions caused by hidden nodes.

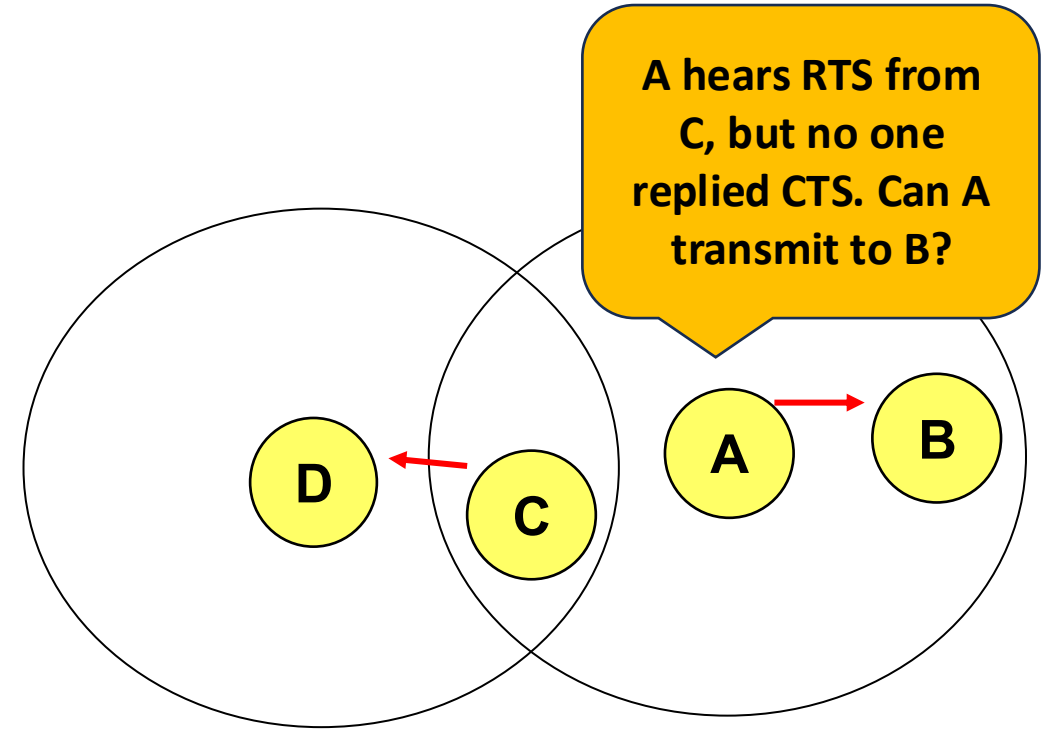
Exposed Terminal Problem



- Exposed terminals (A valid comm can not take place because the sender is exposed)
 - A starts sending to B.
 - C senses carrier, finds medium in use and has to wait for A->B to end.
 - D is outside the range of A, therefore waiting is not necessary.
 - **A and C are “exposed” terminals** i.e. A and C could communicate to their receivers at the same time (since their receivers are far apart), but because A and C are exposed to each other (i.e. can hear each other), one of them needlessly refrains from transmitting.
- A->B and C->D transmissions could have taken place in parallel without collisions, but CS function will only allow one of them

RTS/CTS to recover Exposed Terminal Problem

- Sense channel (CS)
- If busy
 - Freeze backoff timer
- Else
 - Send RTS
 - If CTS not received
 - Back-off to try again later
 - Else
 - Send Data
 - If ACK not received
 - Back-off to try again later
 - Next packet processing



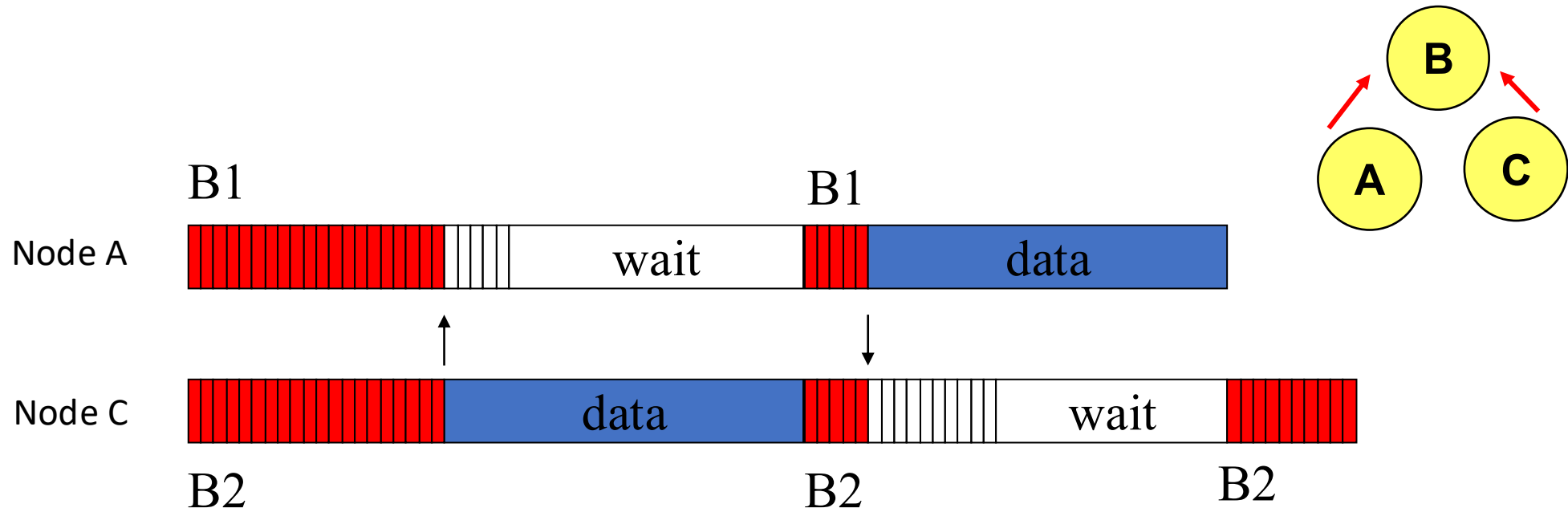
RTS/CTS helps in exposed node problem

When a node **hears an RTS from a neighboring node, but not the corresponding CTS**, that node can deduce that it is an exposed node and is permitted to transmit to other neighboring nodes.

CSMA/CA Algorithm (Contention Window)

- Maintain a value **CW** (Contention Window)
- If Busy,
 - Wait till channel is idle. Then choose a **random** number between 0 and CW and start a back-off timer for proportional amount of time.
 - If transmissions within back-off amount of time, freeze back-off timer and start it once channel becomes idle again.
- If Collisions (Control or Data)
 - Binary exponential increase (doubling) of CW

CSMA/CA Backoff Example



**B1 and B2 are backoff intervals
at nodes A and C**

Backoff Interval

- The time spent counting down backoff intervals is a part of MAC overhead
- Choosing a *large cw* leads to large backoff intervals and can result in larger overhead
- Choosing a *small cw* leads to a larger number of collisions (when two nodes count down to 0 simultaneously)

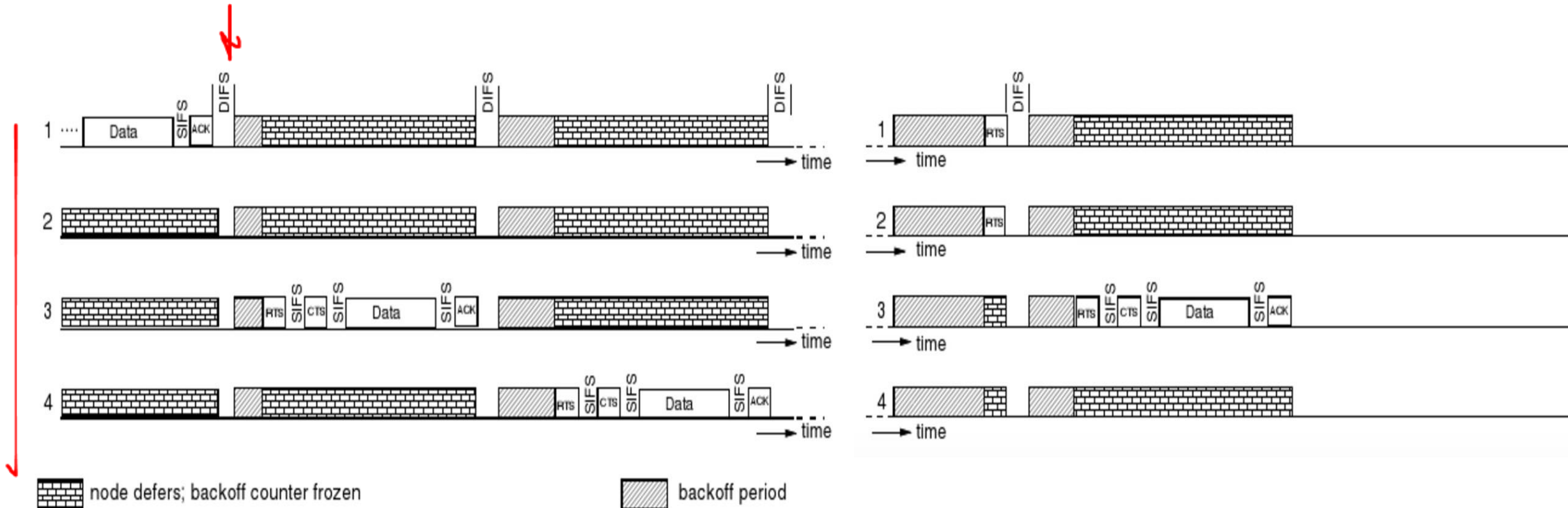
Backoff Interval

- Since the number of nodes attempting to transmit simultaneously may change with time, some mechanism to manage contention is needed
- IEEE 802.11 : contention window *cw* is chosen dynamically depending on collision occurrence

Binary Exponential Backoff in 802.11

- When a node fails to receive CTS in response to its RTS, it increases the contention window
 - *cw* is doubled (up to an upper bound)
- When a node successfully completes a data transfer, it restores *cw* to *Cwmin*

Example scenario: 802.11 DCF exchange



Problem (similar to the HW2 problem)

In the 802.11 DCF timing diagram (see last slide), suppose the following frame spacing and transmission durations are given:

DIFS = 12

SIFS = 7

ACK = RTS = CTS = 15

DATA = 80

Propagation delay = 5

In case of a collision, nodes involved in the collision deterministically double their initial timer value.

Assume there are two nodes, A and B. The initial condition is at $t=0$ where the channel has been idle for a long time, Node A timer = 5, Node B timer = 8, and each of the nodes has data to send, of duration DATA. The time until both nodes have completed received the data (i.e., completed receiving the latest of the two ACK packets) is _____?

Additional slides

Pure ALOHA efficiency

$$\begin{aligned} P(\text{success by given node}) &= P(\text{node transmits}) * \\ &\quad P(\text{no other node transmits in } [t_0-1, t_0]) * \\ &\quad P(\text{no other node transmits in } [t_0, t_0+1]) \\ &= p \cdot (1-p)^{N-1} \cdot (1-p)^{N-1} \\ &= p \cdot (1-p)^{2(N-1)} \end{aligned}$$

$$\begin{aligned} \dots \text{ choosing optimum } p \text{ and then letting } n \rightarrow \infty \\ = 1/(2e) = .18 \end{aligned}$$

even worse than slotted Aloha!

Additional slides

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